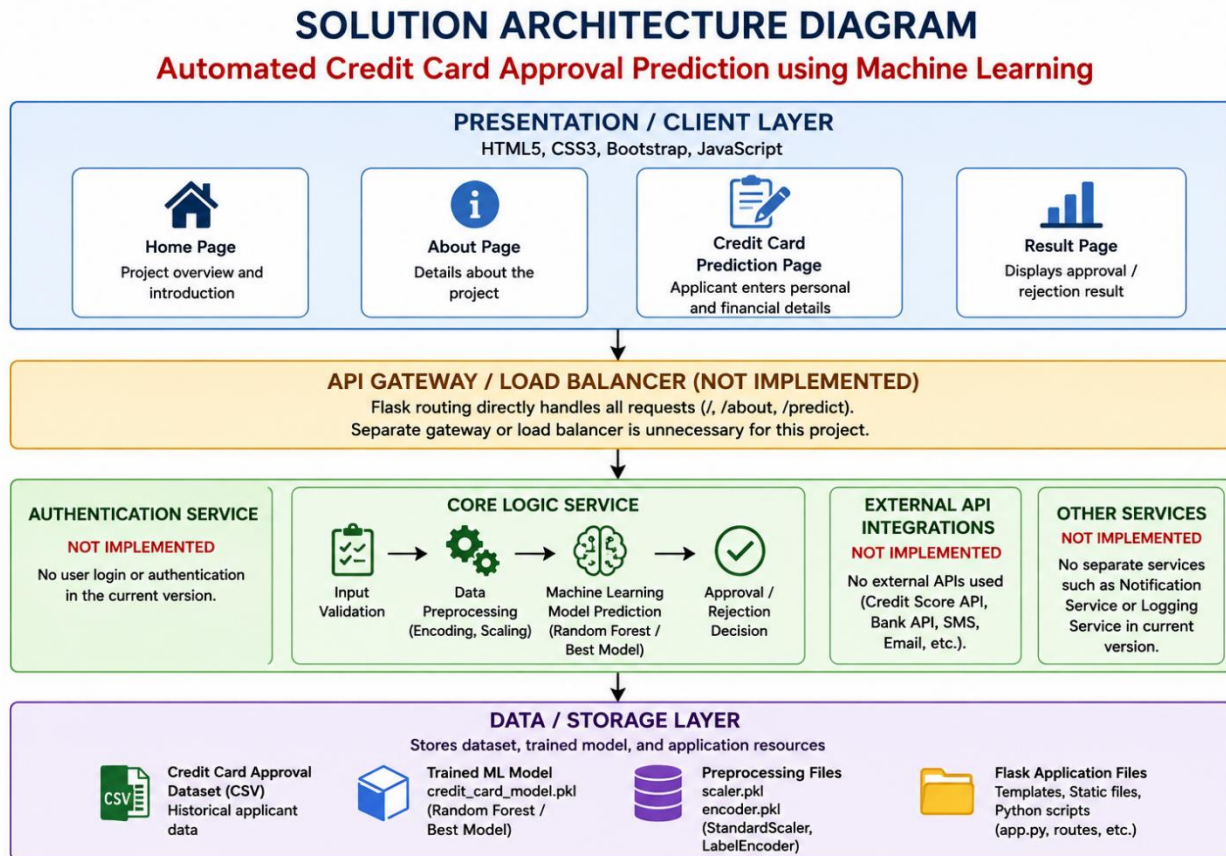


Solution Architecture

Date	08-06-2026
Team ID	
Project Name	Automated Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Solution Architecture Diagram:



This diagram maps credit card approval actual architecture onto the Presentation, Application, and Data layer structure. Layers that are not part of the current implementation (API Gateway, Auth Service, External API Integrations) are explicitly marked as not implemented, rather than filled in with technologies that were not actually used.

Component Description Table



Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Displays the Home, About, Credit Card Prediction, and Result pages. Collects applicant information such as age, income, employment status, credit history, loan amount, and existing debt through a web form and displays the prediction result.	HTML5, CSS3 (custom-styled, no JS framework)
API Gateway	Not Implemented. Flask routing directly handles all application requests (/ , /about, /predict). A separate API Gateway or Load Balancer is unnecessary for this project.	<i>Not applicable</i>
Database	Stores the training dataset, trained Machine Learning model, preprocessing files, and application resources. The trained model is loaded into memory during application startup.	CSV file (Crop_recommendation.csv), Pickle (crop_model.pkl, scaler.pkl)
Core Logic / Microservices	Flask application validates applicant data, preprocesses input using Label Encoding and StandardScaler, loads the trained Machine Learning model, and predicts whether the credit card application is Approved or Rejected.	Python, Flask, Scikit-learn (LogisticRegression, KMeans, StandardScaler), NumPy